MIL-T-52296C
13 October 1983
SUPERSEDING
MIL-T-52296B
16 February 1970

MILITARY SPECIFICATION

TRANSOM ASSEMBLY: 60-TON UNIVERSAL TRANSOM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a transom assembly and its components to be used with a trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood, Open, 12,000 and 16,000 Pound Capacity.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13207E9701	- Transom Assembly.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A5.1	- Carbon Steel Covered Arc Welding Electrodes.
ANSI/AWS A5.10	- Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes.
ANSI/AWS A5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
ANSI/AWS D1.1	- Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The transom assembly and its components shall be as shown on Top Assembly TA13207E9701 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the transom assembly. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Material shall be new and unused.

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3.3.1 Material deterioration and control. The transom assembly shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Welding electrodes.

3.4.1 For steel. Welding electrodes for steel shall conform to ANSI/AWS A5.1 or A5.18, classification E60 or E70 series, size as required.

3.4.2 For aluminum. Welding electrodes for aluminum shall conform to ANSI/AWS A5.10, classification ER4043, size as required.

3.5 Identification marking. The transom assembly or components, as applicable, shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The transom assembly or components, as applicable, except pins, bolts, and threaded portions of threaded components and parts shall be cleaned, treated, and painted in accordance with MIL-T-704, type 8. The contact surfaces of all dissimilar metals shall be painted prior to assembly.

3.7 Stenciling. The nomenclature, National stock number, weight, and cube shall be stenciled on each transom assembly. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

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3.8 Workmanship.

3.8.1 Steel fabrication. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. All bends shall be made by use of metal dies or fixtures to insure uniformity in size and shape. All burrs shall be removed.

3.8.2 Aluminum fabrication. Aluminum used in the fabrication of the transom assembly shall be free from kinks and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Surfaces and edges of all finished members shall be smooth and free from kinks, bends, and winds. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum shall not be cut by methods using heat. Aluminum less than 3/8 inch thick may be cut by shearing; aluminum 3/8 inch thick and over shall be cut by routing or by high-speed saws. Shearing and sawing shall be done neatly and accurately.

3.8.3 Bolted connections. Bolt holes shall be punched or drilled, and all burrs shall be removed.

3.8.4 Riveted connections.

3.8.4.1 Rivet holes. Holes for countersunk rivets shall be of such depth and angle that the rivet head shall fill the hole without projection or recession.

3.8.4.2 Riveting. Components shall be bolted in position before riveting. Jigs or frames shall be used to maintain alinement and tolerance. Countersunk rivets shall be flush with the surrounding surface. Heads of driven rivets shall be uniform in shape and concentric with the rivet shank axis, and shall have full bearing on the surfaces of the joined members. Rivets shall fill the holes in the joined members. Caulking or redriving of rivet heads shall not be permitted. Rivet tension shall not be created by driving in incorrectly fitted members. A rivet with a head deficient in size, not uniform in shape, off-center, loose, burned, or misshapen in any way shall be considered defective. Defective rivets shall be removed without injury to the joined metal. Rivets shall not be heated in excess of 950° F and shall be driven at a temperature of not less than 850° F. Rivets may be driven cold, immediately after quenching, in which case a hole diameter 1/64 inch smaller than for a hot-driven rivet of the same size is required.

3.8.5 Welding and welders.

3.8.5.1 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the ANSI/AWS D1.1 Structural Welding Code - Steel. All welded

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parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

3.8.5.1.1 Steel welding. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed.

3.8.5.1.2 Aluminum welding. Aluminum members to be welded may be preheated for welding to a temperature of not more than 400° F. Aluminum welds may be filed, wirebrushed, or chipped but shall not be hammered.

3.8.5.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ASME or AWS codes.

3.8.6 Interchangeability. Transom assemblies shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article transom assembly or component(s), as applicable, shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each transom assembly shall be inspected for conformance to the dimensions covered by the Quality assurance provisions (QAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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4.5 Inspection procedure.

4.5.1 Examination. The transom assembly or components, as applicable, shall be examined as specified herein for the following defects:

Major

101. All dimensions designated major on the QAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and/or operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Components missing or not as specified.
107. Workmanship not as specified.
108. Rivets not as specified.
109. Weld defects such as flux inclusions, spatter, under-cutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor

201. All dimensions not designated major on the QAP not as specified.
202. Identification marking incorrect, missing, or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The preproduction pack of crated transom assemblies for level A or B shall be subjected to the rail-impact test as specified in MIL-STD-1186, appendix A. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

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4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Containers not as specified for level A or B.	Each incorrect container shall be considered one defect.
111.	Blocking, bracing, and anchoring not in accordance with the referenced document for level A or B.	
112.	Loose nuts, bolts, screws, and washers not secured to the item for which intended for level A or B.	
113.	Components procured separately not secured within the box in accordance with the referenced document for level A or B.	
114.	Gross weight or size exceeds the limitations of the fiberboard box for level B.	
115.	Fiberboard box not sealed and reinforced as specified for level B.	
116.	Packing not in accordance with the referenced document as specified for commercial.	
117.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.	

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination and test within the time frame specified, to prove prior to starting production packaging that the applied packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the first article item or production item. If the first article item is utilized, any packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article item and the production items.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each completely assembled transom assembly (with pins in place) shall be packed in a crate conforming to MIL-C-3774, nailed assembly, skid-type base. Blocking, bracing, and anchoring shall be in accordance with MIL-STD-1186 and the appendix to the crate specification.

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5.2.1.1 Components procured separately from the transom assembly. When the transom latch, shackle assembly, spring holder assembly, transom pin, transom latch pin, and pin adapter are procured separately from the transom assembly, the items of like description shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Loose nuts, bolts, screws, or washers involved shall be secured to the item for which intended to prevent their loss. The components shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.2 Level B. Each completely assembled transom assembly (with pins in place) shall be packed as specified in 5.2.1.

5.2.2.1 Components procured separately from the transom assembly. When the transom latch, shackle assembly, spring holder assembly, transom pin, transom latch pin, and pin adapter are procured separately from the transom assembly, the items of like description shall be packed together in a close-fitting box conforming to PPP-B-601, domestic type, style optional or PPP-B-636, V3c. Loose bolts, nuts, screws, or washers involved shall be secured to the item for which intended to prevent their loss. The components shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The gross weight or size of fiberboard boxes shall not exceed the limitations of the box specification, and the boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification.

5.2.3 Commercial. Each completely assembled transom assembly (with pins in place) shall be shipped unboxed (loose). Components procured separately from the transom assembly shall be packed in accordance with MIL-STD-1188.

5.3 Marking (see 6.7).

5.3.1 Military. Shipping containers and unboxed transom assemblies for level A or B shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The 60-ton universal transom assembly is a component of a trestle assembly to be used in the assembly of fixed and floating bridges and in reinforcing existing fixed bridges.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. When a preproduction pack is not required (see 5.1).

d. Time frame required for submission of the preproduction pack (see 5.1).

e. Level of packing required (see 5.2).

6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced transom assembly. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the documents first article.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to pack and mark the transom assembly or separately procured components in accordance with this specification.

6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.6 Definitions.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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6.7 Packaging. Preservation is not required, but for purposes of preservation/packing level marking, preservation shall be designated as level A.

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Review activity:
Navy - MC

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-52296C	Transom Assembly: 60-Ton Universal Transom

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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